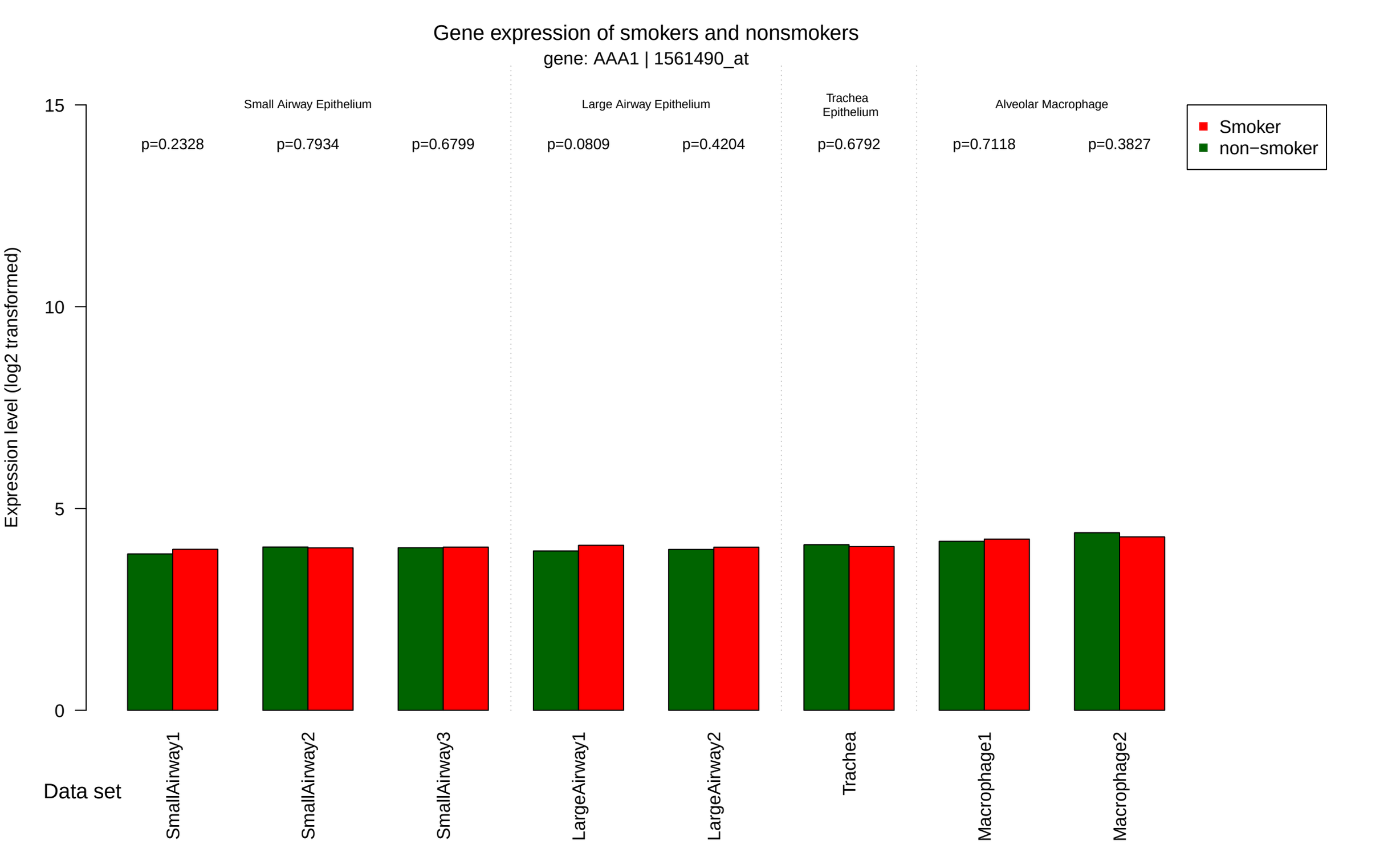
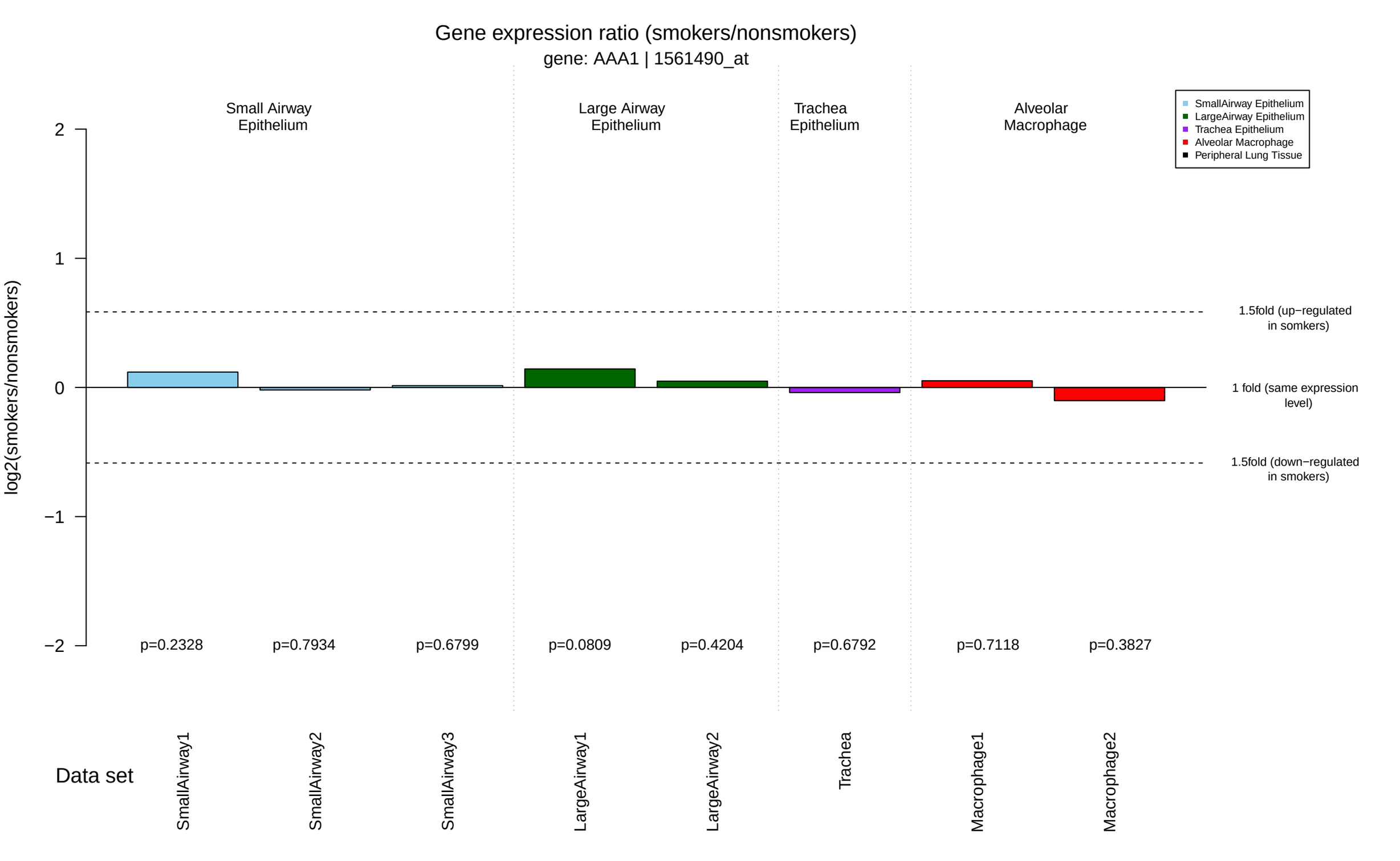
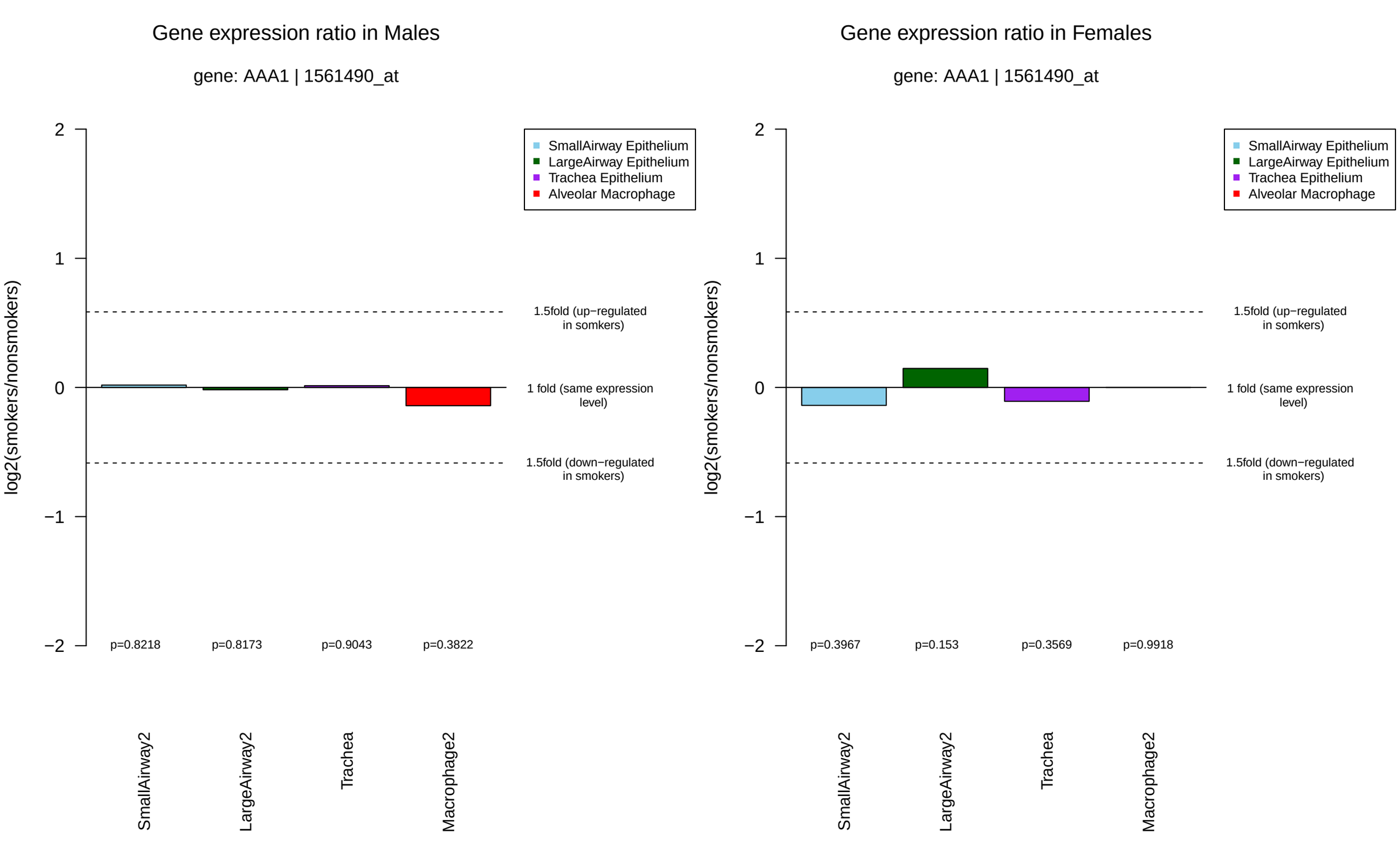
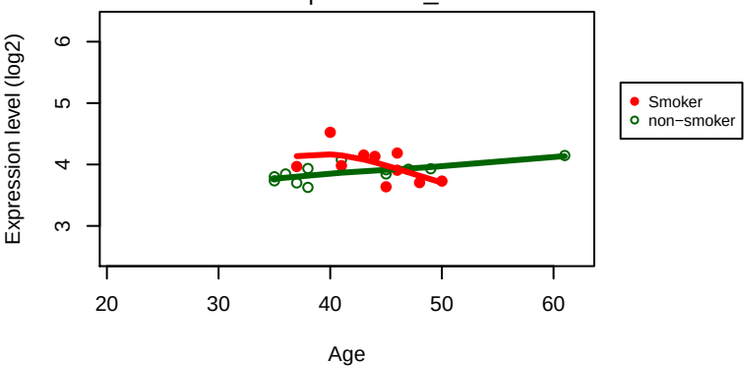
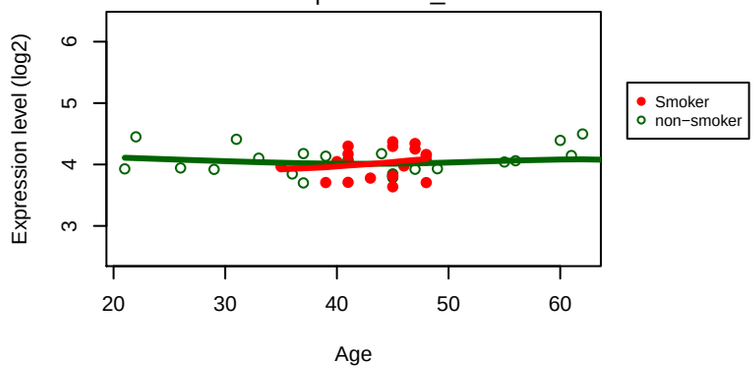
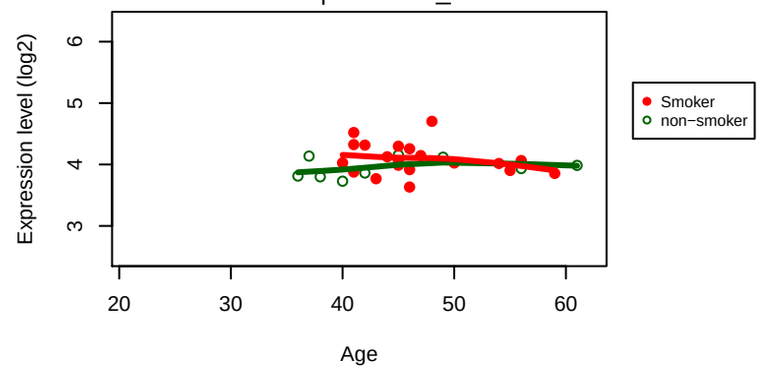

Gene expression of smokers and nonsmokers
gene: AAA1 | 1561490_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2328
p=0.7934
p=0.6799
p=0.0809
p=0.4204
p=0.6792
p=0.7118
p=0.3827
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

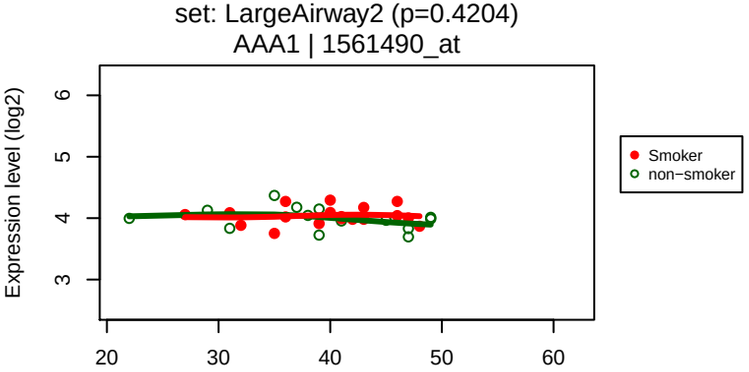

Gene expression ratio (smokers/nonsmokers)
gene: AAA1 | 1561490_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2328
p=0.7934
p=0.6799
p=0.0809
p=0.4204
p=0.6792
p=0.7118
p=0.3827
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

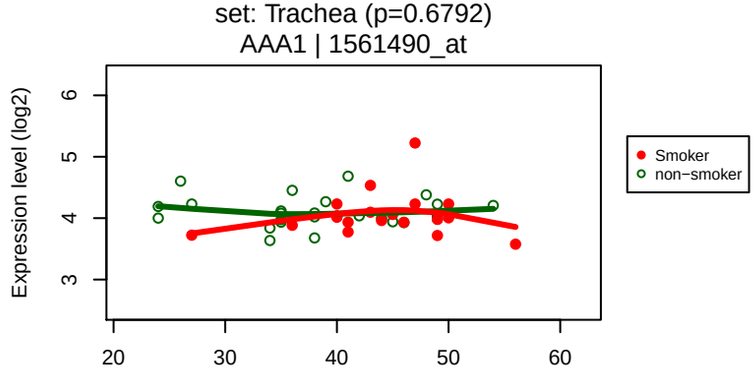

Gene expression ratio in Males
gene: AAA1 | 1561490_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8218
p=0.8173
p=0.9043
p=0.3822
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AAA1 | 1561490_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3967
p=0.153
p=0.3569
p=0.9918
SmallAirway2
LargeAirway2
Trachea
Macrophage2

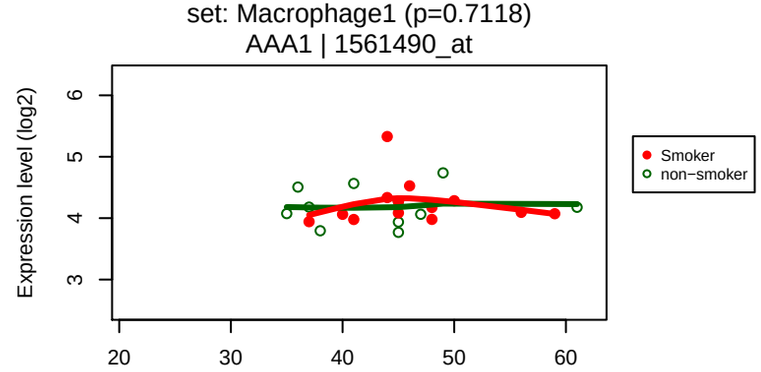

set: SmallAirway1 (p=0.2328)
AAA1 | 1561490_at
Expression level (log2)
Age
Smoker
non−smoker

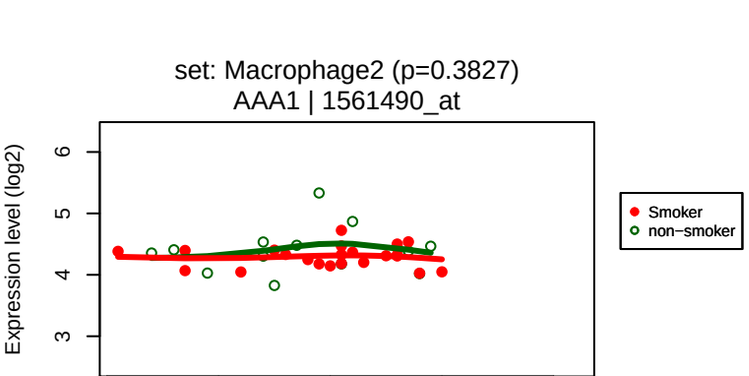

set: SmallAirway2 (p=0.7934)
AAA1 | 1561490_at
Expression level (log2)
Age
Smoker
non−smoker

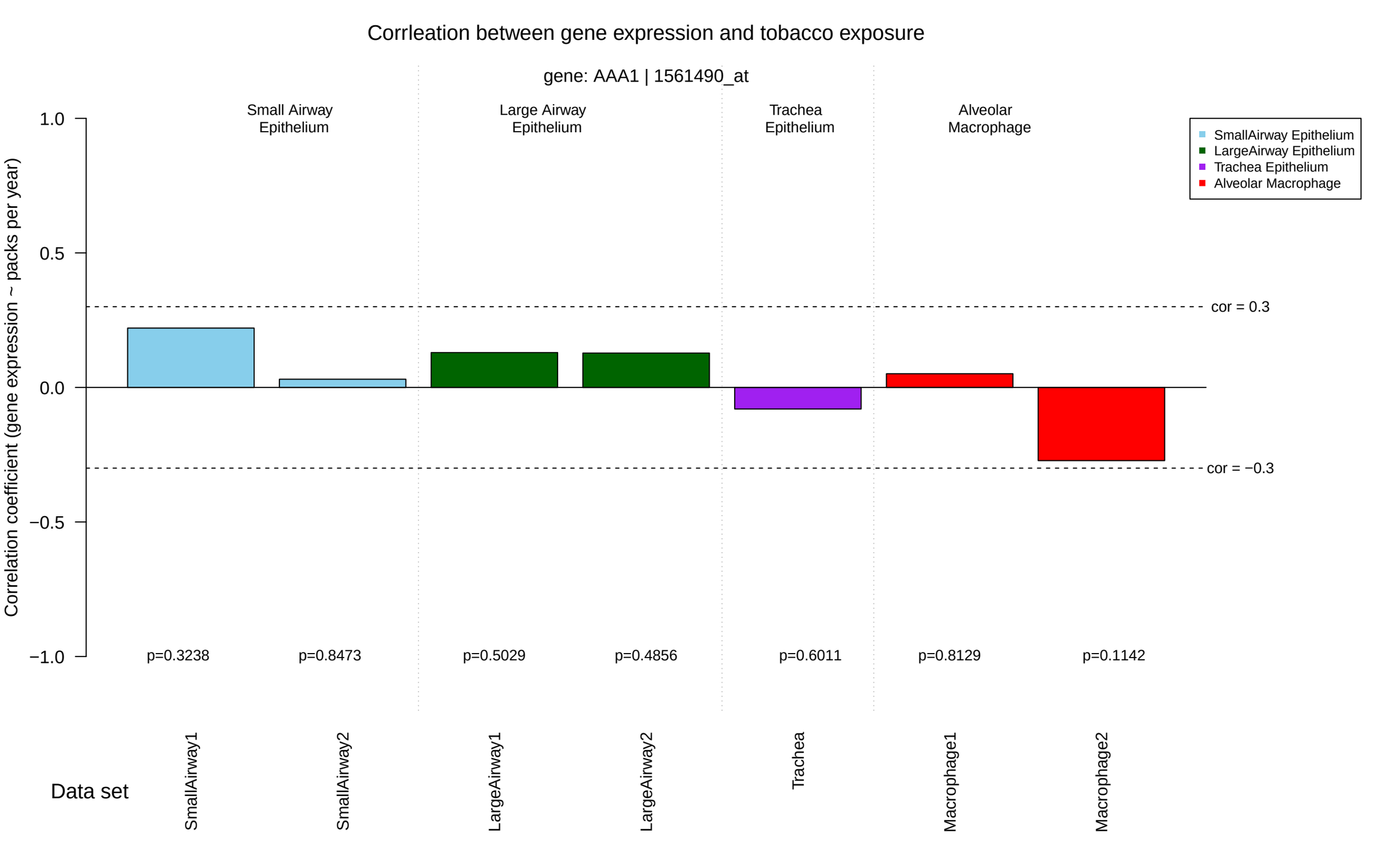

set: LargeAirway1 (p=0.0809)
AAA1 | 1561490_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4204)
AAA1 | 1561490_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6792)
AAA1 | 1561490_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7118)
AAA1 | 1561490_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3827)
AAA1 | 1561490_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AAA1 | 1561490_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3238
p=0.8473
p=0.5029
p=0.4856
p=0.6011
p=0.8129
p=0.1142
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2